

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 State and explain Bernoulli equation. How the temperature and pressure vary w.r.t altitude?
- Q.24 Describe the various control surfaces and their working and the effect on the movement of the aircraft.
- Q.25 Describe the working of hydraulic system with the function of each component explained.

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1st Sem / Aeronautical Engineering Subject:- Basics of Aeronautics

Time : 3Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 The lift and drag coefficients are not primarily a function of_____
- a) vehicle configuration
 - b) flight Mach number
 - c) angle of attach
 - d) vehicle mass
- Q.2 The cross-sectional shape of wing is called_____
- a) airfoil
 - b) circle
 - c) camber
 - d) chord
- Q.3 If an airfoil is operating at Mach 1.1, then it is working in_____
- a) transonic range
 - b) supersonic range
 - c) subsonic range
 - d) hypersonic range

- Q.4 What is the basic assumption taken in formulating equations of motion?
- All the engines will operate at equal gross thrust
 - All the engines do not operate at equal gross thrust
 - All the engines will operate at equal gross drag
 - All the engines do not operate at equal gross drag
- Q.5 If the aircraft has a power-producing engine which have driven propellers then the power is to be converted to thrust to formulate the equations of motion.
- True
 - False
- Q.6 What is a chord?
- Distance between leading edge and chord
 - Distance between chord and chamber
 - Distance between leading edge and trailing edge
 - Distance between trailing edge and chord

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 What is the temperature of outside an airliner flying at 10Km height?

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- Q.8 What is a pure monocoque structure?
- Q.9 What is the use of hydraulic system in an aircraft?
- Q.10 What is the special about rocket engine?
- Q.11 What is a transonic range of speed?
- Q.12 Turbojet is more/less efficient as compared to turbofan engine.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What is the difference between a turbojet and turbofan engine?
- Q.14 What are the structural components of fuselage?
- Q.15 What is the function of flaps in the airplane?
- Q.16 What is an airfoil and its use?
- Q.17 Draw the forces on an airplane in climbing attitude.
- Q.18 Explain any one pneumatic system of aircraft.
- Q.19 What are various types of control systems?
- Q.20 What are various control surfaces in an airplane?
- Q.21 How the helicopter motion is controlled?
- Q.22 What are the non-air breathing engines used?

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